Conceptual question · Unit 2 · 2 points · 15 min · Conceptual / fast

C07 — Damped and forced oscillator: energy decay per period and where the amplitude really peaks

A block of mass $m = 2$ kg moves along the x axis of an inertial frame, attached to a spring of stiffness $k = 50$ N/m and to a viscous damper of constant $c = 4$ N s/m. Its equation of motion is $m\ddot{x} + c\dot{x} + kx = F(t)$, and (Notes Eq. 5.23) $\gamma = c/(2m)$, $\omega_0 = \sqrt{k/m}$. Decide whether each statement is TRUE or FALSE and justify the answer. A verdict without a justification scores nothing.

Part (a) · 1 point

Free motion

With $F(t) = 0$. Statement: "The motion is underdamped, and over each period $T = 2\pi/\omega$ of the damped oscillation the mechanical energy $E = \frac{1}{2}m\dot{x}^2 + \frac{1}{2}kx^2$ is multiplied by the factor $e^{-2\pi/\sqrt{6}} \approx 0.0769$, the square of the factor by which the amplitude decays."

Step 1. Type of motion

$\gamma = \dfrac{c}{2m} = \dfrac{4}{4} = 1\ \text{s}^{-1}$ and $\omega_0 = \sqrt{50/2} = 5\ \text{s}^{-1}$. The roots of $mp^2 + cp + k = 0$ are $p = -\gamma \pm \sqrt{\gamma^2 - \omega_0^2} = -1 \pm 2\sqrt{6}\,i$: complex, so the motion is **underdamped** (Notes §5.3.1), with $\omega = \sqrt{\omega_0^2 - \gamma^2} = \sqrt{24} = 2\sqrt{6}\ \text{s}^{-1}$ and $T = 2\pi/\omega \approx 1.283$ s.

Step 2. Exact decay over one period

The general solution is $x = e^{-\gamma t}\big(A\cos\omega t + B\sin\omega t\big)$. Since cos and sin have period $T = 2\pi/\omega$, $x(t+T) = e^{-\gamma(t+T)}\big(A\cos\omega t + B\sin\omega t\big) = e^{-\gamma T}x(t)$, and differentiating this identity, $\dot{x}(t+T) = e^{-\gamma T}\dot{x}(t)$. Both terms of E are quadratic:

$$E(t+T) = e^{-2\gamma T}E(t), \qquad \gamma T = \frac{2\pi}{2\sqrt{6}} = \frac{\pi}{\sqrt{6}}$$

Step 3. Numbers

Amplitude factor $e^{-\pi/\sqrt{6}} \approx 0.277$; energy factor $e^{-2\pi/\sqrt{6}} = \big(e^{-\pi/\sqrt{6}}\big)^2 \approx 0.0769$. The energy is dissipated by the damper: $\dot{E} = \dot{x}\,(m\ddot{x} + kx) = -c\dot{x}^2 \le 0$. The statement is TRUE.

KEY POINTS

- TRUE.
- $\gamma = 1\ \text{s}^{-1} < \omega_0 = 5\ \text{s}^{-1}$: underdamped; $\omega = \sqrt{\omega_0^2 - \gamma^2} = 2\sqrt{6} \approx 4.899\ \text{s}^{-1}$, $T \approx 1.283$ s.
- $x(t+T) = e^{-\gamma T}x(t)$ and $\dot{x}(t+T) = e^{-\gamma T}\dot{x}(t)$ exactly, with $\gamma T = \pi/\sqrt{6}$: amplitude factor $e^{-\pi/\sqrt{6}} \approx 0.277$, energy factor $e^{-2\pi/\sqrt{6}} \approx 0.0769$.

Part (b) · 1 point

Forced motion

Now $F(t) = F_1 \cos \omega_1 t$ with $F_1 = 10$ N. Statement: "In the steady state the amplitude is largest when the forcing frequency equals $\omega_0 = 5\ s^{-1}$; there the response lags the force by 90° and its amplitude is 0.5 m."

Step 1. Steady response at $\omega_1 = \omega_0$

After the transient dies out (Notes §5.4), $x = a_1 \cos(\omega_1 t + \alpha_1)$ with a_1 above and $\tan \alpha_1 = -\dfrac{2\gamma\omega_1}{\omega_0^2 - \omega_1^2}$ (Notes Eq. 5.42). At $\omega_1 = \omega_0$ the denominator of $\tan \alpha_1$ vanishes and $\alpha_1 = -\pi/2$: a lag of 90°. The amplitude is

$$a_1(\omega_0) = \frac{F_1/m}{2\gamma\omega_0} = \frac{10/2}{2 \cdot 1 \cdot 5} = 0.5 \text{ m}$$

Step 2. Where the maximum over ω_1 is

Let $s = \omega_1^2$ and $D(s) = (\omega_0^2 - s)^2 + 4\gamma^2 s$. a_1 is largest where D is smallest:

$$\frac{dD}{ds} = -2(\omega_0^2 - s) + 4\gamma^2 = 0 \implies s^* = \omega_0^2 - 2\gamma^2 = 25 - 2 = 23 \text{ s}^{-2}$$

($d^2D/ds^2 = 2 > 0$: a minimum). Substituting, $D(s^*) = 4\gamma^4 + 4\gamma^2(\omega_0^2 - 2\gamma^2) = 4\gamma^2(\omega_0^2 - \gamma^2)$, so

$$a_{1,max} = \frac{F_1/m}{2\gamma\sqrt{\omega_0^2 - \gamma^2}} = \frac{5}{2\sqrt{24}} \approx 0.510 \text{ m} > 0.5 \text{ m}$$

at $\omega_1 = \sqrt{23} \approx 4.796$ s^{-1}, where $\tan \alpha_1 = -\dfrac{2\sqrt{23}}{2} = -\sqrt{23}$, $\alpha_1 \approx -78.2^\circ$.

Step 3. Verdict

The result "maximum at ω_0" of Notes Eq. 5.45 holds when ω_1 is fixed and ω_0 is varied; when the forcing frequency is varied, the peak is shifted below ω_0 by the damping. The statement is FALSE.

KEY POINTS

- FALSE (the phase and the value at $\omega_1 = \omega_0$ are right; the location of the maximum is not).
- At $\omega_1 = \omega_0$: $a_1 = \dfrac{F_1}{2m\gamma\omega_0} = 0.5$ m, $\tan\alpha_1 \to -\infty$, $\alpha_1 = -90^\circ$.
- Maximum over ω_1 at $\omega_1^2 = \omega_0^2 - 2\gamma^2 = 23$ s^{-2} ($\omega_1 \approx 4.796$ s^{-1}): $a_{1,max} = \dfrac{F_1}{2m\gamma\sqrt{\omega_0^2 - \gamma^2}} = \dfrac{5}{2\sqrt{24}} \approx 0.510$ m > 0.5 m, with $\alpha_1 \approx -78.2^\circ$.

Interpretation and checks

- **Dimensional check:** $\gamma, \omega_0, \omega_1$ are s^{-1}; F_1/m is m s^{-2} and $(F_1/m)/(\gamma\omega_0)$ is m.
- **Limits:** for $\gamma \to 0$ the peak returns to ω_0 and $a_{1,max} \to \infty$; for $\gamma \geq \omega_0/\sqrt{2}$ there is no peak at all ($s^* \leq 0$) and a_1 decreases from its static value $F_1/k = 0.2$ m.
- **Common mistakes in the exam:** (1) using the same decay factor for the amplitude and for the energy; (2) placing the amplitude peak at ω_0 for any damping; (3) confusing the 90° phase lag (exactly at ω_0) with the amplitude maximum.